Phase 1 - Data Prep

In [ ]:
```
import pandas as pd
import numpy as np

def load_data(url):
    # Column names as defined by UCI dataset documentation
    column_names = [
        'id',
        'clump_thickness',
        'cell_size_uniformity',
        'cell_shape_uniformity',
        'marginal_adhesion',
        'single_ep_cell_size',
        'bare_nuclei',
        'bland_chromatin',
        'normal_nucleoli',
        'mitoses',
        'class'
    ]

    # Load the CSV — no header exists in the file so header=None
    df = pd.read_csv(url, header=None, names=column_names)

    # Drop the ID column — it carries no useful information
    df = df.drop(columns=['id'])

    return df


#Run it
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/breast-cancer-wisc
df = load_data(url)

# Quick sanity check
print("Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
print("\nData types:")
print(df.dtypes)
```

```
Shape: (699, 10)

First 5 rows:
   clump_thickness  cell_size_uniformity  cell_shape_uniformity  \
0                5                     1                      1
1                5                     4                      4
2                3                     1                      1
3                6                     8                      8
4                4                     1                      1

   marginal_adhesion  single_ep_cell_size bare_nuclei  bland_chromatin  \
0                  1                    2           1                3
1                  5                    7          10                3
2                  1                    2           2                3
3                  1                    3           4                3
4                  3                    2           1                3

   normal_nucleoli  mitoses  class
0                1        1      2
1                2        1      2
2                1        1      2
3                7        1      2
4                1        1      2

Data types:
clump_thickness           int64
cell_size_uniformity      int64
cell_shape_uniformity     int64
marginal_adhesion         int64
single_ep_cell_size       int64
bare_nuclei              object
bland_chromatin           int64
normal_nucleoli           int64
mitoses                   int64
class                     int64
dtype: object
```

In [2]:

```python
def handle_missing_values(df):
    # Step 1 — Replace '?' strings with NaN so pandas can recognize them
    df['bare_nuclei'] = df['bare_nuclei'].replace('?', np.nan)

    # Step 2 — Convert column to numeric (was string/object)
    df['bare_nuclei'] = pd.to_numeric(df['bare_nuclei'])

    # Step 3 — Report how many are missing before filling
    missing_count = df['bare_nuclei'].isna().sum()
    print(f"Missing values in 'bare_nuclei': {missing_count}")

    # Step 4 — Fill missing values with the median of the column
    median_val = df['bare_nuclei'].median()
    print(f"Filling with median value: {median_val}")
    df['bare_nuclei'] = df['bare_nuclei'].fillna(median_val)

    # Step 5 — Convert back to int to match other columns
    df['bare_nuclei'] = df['bare_nuclei'].astype(int)
```

```
    # Step 6 — Final check: any missing values left anywhere?
    print(f"\nTotal missing values in entire dataset: {df.isna().sum().sum()}")

    return df


#  Run it
df = handle_missing_values(df)

# Confirm the fix
print("\nData types after fix:")
print(df.dtypes)
```

```
Missing values in 'bare_nuclei': 16
Filling with median value: 1.0

Total missing values in entire dataset: 0

Data types after fix:
clump_thickness          int64
cell_size_uniformity     int64
cell_shape_uniformity    int64
marginal_adhesion        int64
single_ep_cell_size      int64
bare_nuclei              int64
bland_chromatin          int64
normal_nucleoli          int64
mitoses                  int64
class                    int64
dtype: object
```

In [3]:

```
def normalize_features(df):
    # Separate features from the label
    # We never touch the class column during normalization
    feature_cols = [col for col in df.columns if col != 'class']

    # Extract features and labels as separate numpy arrays
    X = df[feature_cols].values.astype(float)  # shape: (699, 9)
    y = df['class'].values                      # shape: (699,)

    # Min-Max normalization — done manually (no sklearn!)
    X_min = X.min(axis=0)   # minimum per feature column
    X_max = X.max(axis=0)   # maximum per feature column

    X_normalized = (X - X_min) / (X_max - X_min)

    print("Normalization complete!")
    print(f"Feature matrix shape : {X_normalized.shape}")
    print(f"Labels shape         : {y.shape}")
    print(f"\nBefore normalization (first row): {X[0]}")
    print(f"After  normalization (first row): {X_normalized[0].round(4)}")
    print(f"\nMin per feature: {X_normalized.min(axis=0)}")
    print(f"Max per feature: {X_normalized.max(axis=0)}")

    return X_normalized, y
```

```
# Run it
X, y = normalize_features(df)
```

```
Normalization complete!
Feature matrix shape : (699, 9)
Labels shape         : (699,)

Before normalization (first row): [5. 1. 1. 1. 2. 1. 3. 1. 1.]
After  normalization (first row): [0.4444 0.     0.     0.     0.1111 0.     0.2222
0.     0.    ]

Min per feature: [0. 0. 0. 0. 0. 0. 0. 0. 0.]
Max per feature: [1. 1. 1. 1. 1. 1. 1. 1. 1.]
```

Phase 2 — K-Means

How it works

1. Pick K random points from the data as starting centroids
2. Assign every point to the nearest centroid using Euclidean distance
3. Recalculate each centroid as the mean of all points in its cluster
4. Repeat steps 2-3 until the centroids stop moving (convergence)

How we picked K

We used the Elbow Method , we ran K-Means for K = 1 to 10 and plotted the WCSS for each K. The WCSS dropped sharply from K=1 to K=2 and then slowed down, so K=2 was the clear choice. This also makes sense because the dataset has exactly two classes: Benign and Malignant.

In [4]:
```
def initialize_centroids(X, k):

    # Randomly pick k data points from X as starting centroids.
    # We use a fixed random seed so results are reproducible.

    np.random.seed(42)

    # Randomly choose k indices from the dataset (no replacement)
    indices = np.random.choice(X.shape[0], size=k, replace=False)

    # Return those rows as our initial centroids
    centroids = X[indices]

    print(f"Initialized {k} centroids from random data points")
    print(f"Centroids shape: {centroids.shape}")

    return centroids
```

```
# Run it
centroids = initialize_centroids(X, k=2)
print(f"\nFirst centroid : {centroids[0].round(4)}")
print(f"Second centroid: {centroids[1].round(4)}")
```

```
Initialized 2 centroids from random data points
Centroids shape: (2, 9)

First centroid : [0.     0.     0.1111 0.     0.2222 0.     0.     0.     0.    ]
Second centroid: [0.3333 0.     0.     0.     0.1111 0.     0.1111 0.     0.    ]
```

In [5]:

```
def assign_clusters(X, centroids):

    distances = []

    for centroid in centroids:
        # calculate euclidean distance from every point to this centroid
        diff = X - centroid
        dist = np.sqrt((diff ** 2).sum(axis=1))
        distances.append(dist)

    # distances is now shape (k, 699), we transpose to get (699, k)
    distances = np.array(distances).T

    # assign each point to the closest centroid
    labels = np.argmin(distances, axis=1)

    return labels


# run it
labels = assign_clusters(X, centroids)

print("Labels shape:", labels.shape)
print("Unique clusters:", np.unique(labels))
print("Points in cluster 0:", (labels == 0).sum())
print("Points in cluster 1:", (labels == 1).sum())
print("First 20 labels:", labels[:20])
```

```
Labels shape: (699,)
Unique clusters: [0 1]
Points in cluster 0: 210
Points in cluster 1: 489
First 20 labels: [1 1 1 1 1 1 0 0 0 1 0 0 1 0 1 1 1 1 1 1]
```

In [9]:

```
def update_centroids(X, labels, k):

    new_centroids = []

    for i in range(k):
        cluster_points = X[labels == i]

        # if a cluster is empty, reinitialize to a random point
        if len(cluster_points) == 0:
            random_idx = np.random.randint(0, X.shape[0])
            new_centroids.append(X[random_idx])
        else:
```

```
            new_centroids.append(cluster_points.mean(axis=0))

    return np.array(new_centroids)
# run it
new_centroids = update_centroids(X, labels, k=2)

print("New centroids shape:", new_centroids.shape)
print("\nCentroid 0:", new_centroids[0].round(4))
print("Centroid 1:", new_centroids[1].round(4))

# see how much the centroids moved from initialization
print("\nHow much centroids moved:")
print("Centroid 0 shift:", np.round(np.abs(new_centroids[0] - centroids[0]), 4))
print("Centroid 1 shift:", np.round(np.abs(new_centroids[1] - centroids[1]), 4))
```

```
New centroids shape: (2, 9)

Centroid 0: [0.227  0.0335 0.0492 0.0375 0.1209 0.033  0.1226 0.028  0.0122]
Centroid 1: [0.6833 0.642  0.6349 0.5252 0.4953 0.7597 0.5655 0.5641 0.1714]

How much centroids moved:
Centroid 0 shift: [0. 0. 0. 0. 0. 0. 0. 0. 0.]
Centroid 1 shift: [0. 0. 0. 0. 0. 0. 0. 0. 0.]
```

In [10]:

```
def kmeans(X, k, max_iters=100, tol=1e-4):

    # step 1 - initialize centroids
    centroids = initialize_centroids(X, k)

    for i in range(max_iters):

        # step 2 - assign clusters
        labels = assign_clusters(X, centroids)

        # step 3 - update centroids
        new_centroids = update_centroids(X, labels, k)

        # step 4 - check convergence
        shift = np.abs(new_centroids - centroids).sum()

        print(f"Iteration {i+1} | centroid shift: {shift:.6f}")

        if shift < tol:
            print(f"\nConverged at iteration {i+1}!")
            break

        centroids = new_centroids

    return centroids, labels


# run it
centroids, labels = kmeans(X, k=2)

print("\nFinal cluster sizes:")
```

```
for i in range(2):
    print(f"  Cluster {i}: {(labels == i).sum()} points")
```

```
Initialized 2 centroids from random data points
Centroids shape: (2, 9)
Iteration 1 | centroid shift: 2.781981
Iteration 2 | centroid shift: 2.840861
Iteration 3 | centroid shift: 0.370358
Iteration 4 | centroid shift: 0.029592
Iteration 5 | centroid shift: 0.014429
Iteration 6 | centroid shift: 0.000000

Converged at iteration 6!

Final cluster sizes:
  Cluster 0: 465 points
  Cluster 1: 234 points
```

In [11]:

```
import matplotlib.pyplot as plt

def compute_wcss(X, labels, centroids):

    wcss = 0
    for i, centroid in enumerate(centroids):
        cluster_points = X[labels == i]
        wcss += ((cluster_points - centroid) ** 2).sum()

    return wcss


def elbow_method(X, max_k=10):

    wcss_scores = []

    for k in range(1, max_k + 1):
        centroids, labels = kmeans(X, k)
        wcss = compute_wcss(X, labels, centroids)
        wcss_scores.append(wcss)
        print(f"K={k} | WCSS: {wcss:.4f}")

    # plot the elbow curve
    plt.figure(figsize=(8, 5))
    plt.plot(range(1, max_k + 1), wcss_scores, marker='o', color='steelblue', linew
    plt.title('Elbow Method')
    plt.xlabel('Number of Clusters (K)')
    plt.ylabel('WCSS')
    plt.xticks(range(1, max_k + 1))
    plt.grid(True)
    plt.tight_layout()
    plt.show()

    return wcss_scores


# run it
wcss_scores = elbow_method(X)
```

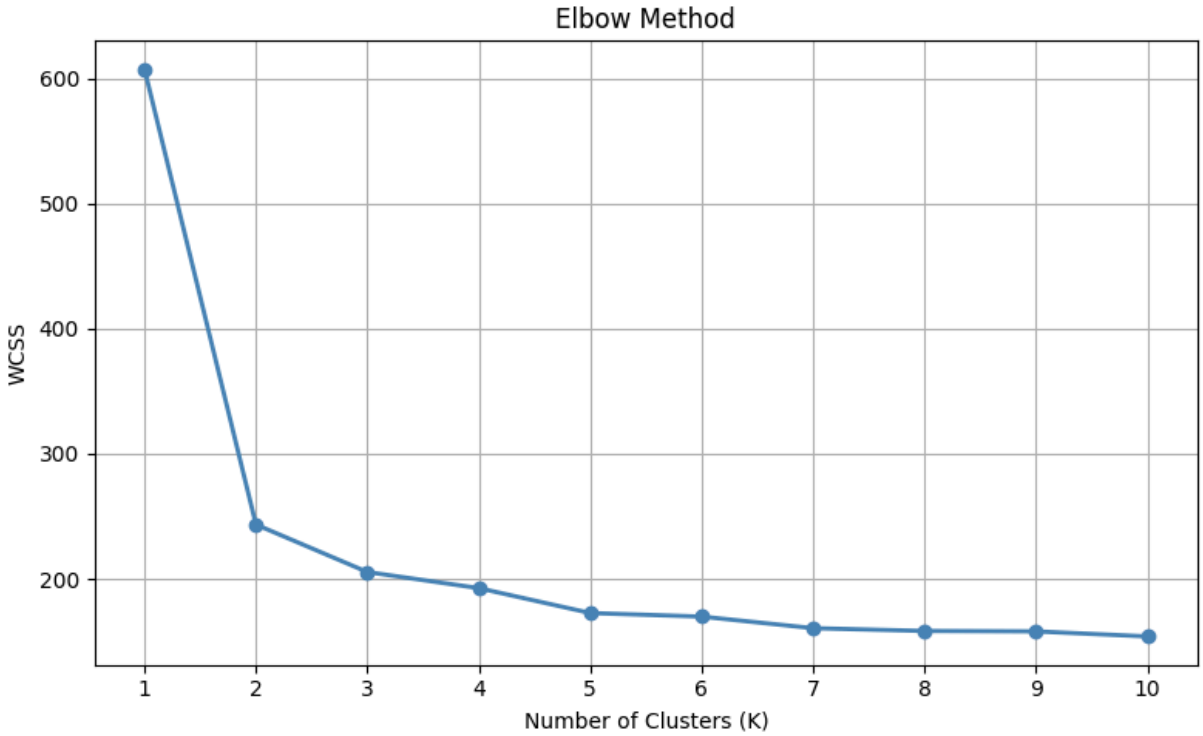
```
Initialized 1 centroids from random data points
Centroids shape: (1, 9)
Iteration 1 | centroid shift: 1.795899
Iteration 2 | centroid shift: 0.000000

Converged at iteration 2!
K=1 | WCSS: 607.1612
Initialized 2 centroids from random data points
Centroids shape: (2, 9)
Iteration 1 | centroid shift: 2.781981
Iteration 2 | centroid shift: 2.840861
Iteration 3 | centroid shift: 0.370358
Iteration 4 | centroid shift: 0.029592
Iteration 5 | centroid shift: 0.014429
Iteration 6 | centroid shift: 0.000000

Converged at iteration 6!
K=2 | WCSS: 243.4369
Initialized 3 centroids from random data points
Centroids shape: (3, 9)
Iteration 1 | centroid shift: 4.632453
Iteration 2 | centroid shift: 3.142753
Iteration 3 | centroid shift: 0.688331
Iteration 4 | centroid shift: 0.635907
Iteration 5 | centroid shift: 0.455454
Iteration 6 | centroid shift: 0.335328
Iteration 7 | centroid shift: 0.156115
Iteration 8 | centroid shift: 0.044732
Iteration 9 | centroid shift: 0.107770
Iteration 10 | centroid shift: 0.107384
Iteration 11 | centroid shift: 0.064841
Iteration 12 | centroid shift: 0.045529
Iteration 13 | centroid shift: 0.053780
Iteration 14 | centroid shift: 0.000000

Converged at iteration 14!
K=3 | WCSS: 205.5144
Initialized 4 centroids from random data points
Centroids shape: (4, 9)
Iteration 1 | centroid shift: 3.227349
Iteration 2 | centroid shift: 0.438002
Iteration 3 | centroid shift: 0.148163
Iteration 4 | centroid shift: 0.082208
Iteration 5 | centroid shift: 0.056401
Iteration 6 | centroid shift: 0.090004
Iteration 7 | centroid shift: 0.153357
Iteration 8 | centroid shift: 0.117645
Iteration 9 | centroid shift: 0.216464
Iteration 10 | centroid shift: 0.492848
Iteration 11 | centroid shift: 0.580778
Iteration 12 | centroid shift: 0.573347
Iteration 13 | centroid shift: 0.644239
Iteration 14 | centroid shift: 0.495866
Iteration 15 | centroid shift: 0.409876
Iteration 16 | centroid shift: 0.280592
Iteration 17 | centroid shift: 0.162593
```

```
Iteration 18 | centroid shift: 0.224768
Iteration 19 | centroid shift: 0.127968
Iteration 20 | centroid shift: 0.113810
Iteration 21 | centroid shift: 0.000000

Converged at iteration 21!
K=4 | WCSS: 192.6545
Initialized 5 centroids from random data points
Centroids shape: (5, 9)
Iteration 1 | centroid shift: 3.894016
Iteration 2 | centroid shift: 0.665900
Iteration 3 | centroid shift: 0.326145
Iteration 4 | centroid shift: 0.125518
Iteration 5 | centroid shift: 0.065576
Iteration 6 | centroid shift: 0.130760
Iteration 7 | centroid shift: 0.174832
Iteration 8 | centroid shift: 0.125903
Iteration 9 | centroid shift: 0.255269
Iteration 10 | centroid shift: 0.512692
Iteration 11 | centroid shift: 0.668201
Iteration 12 | centroid shift: 0.786082
Iteration 13 | centroid shift: 0.745431
Iteration 14 | centroid shift: 0.748772
Iteration 15 | centroid shift: 0.749648
Iteration 16 | centroid shift: 0.727745
Iteration 17 | centroid shift: 0.707047
Iteration 18 | centroid shift: 0.536660
Iteration 19 | centroid shift: 0.466723
Iteration 20 | centroid shift: 0.442126
Iteration 21 | centroid shift: 0.249257
Iteration 22 | centroid shift: 0.239115
Iteration 23 | centroid shift: 0.214710
Iteration 24 | centroid shift: 0.154485
Iteration 25 | centroid shift: 0.169160
Iteration 26 | centroid shift: 0.274671
Iteration 27 | centroid shift: 0.188332
Iteration 28 | centroid shift: 0.230221
Iteration 29 | centroid shift: 0.261241
Iteration 30 | centroid shift: 0.203761
Iteration 31 | centroid shift: 0.129377
Iteration 32 | centroid shift: 0.185811
Iteration 33 | centroid shift: 0.155332
Iteration 34 | centroid shift: 0.060801
Iteration 35 | centroid shift: 0.000000

Converged at iteration 35!
K=5 | WCSS: 172.7611
Initialized 6 centroids from random data points
Centroids shape: (6, 9)
Iteration 1 | centroid shift: 4.011324
Iteration 2 | centroid shift: 0.724537
Iteration 3 | centroid shift: 0.310169
Iteration 4 | centroid shift: 0.395743
Iteration 5 | centroid shift: 0.590938
Iteration 6 | centroid shift: 0.651079
Iteration 7 | centroid shift: 0.547765
```

```
Iteration 8 | centroid shift: 0.733623
Iteration 9 | centroid shift: 0.653803
Iteration 10 | centroid shift: 0.566512
Iteration 11 | centroid shift: 0.431042
Iteration 12 | centroid shift: 0.397999
Iteration 13 | centroid shift: 0.569225
Iteration 14 | centroid shift: 0.763002
Iteration 15 | centroid shift: 0.786763
Iteration 16 | centroid shift: 0.489523
Iteration 17 | centroid shift: 0.591712
Iteration 18 | centroid shift: 0.316460
Iteration 19 | centroid shift: 0.358950
Iteration 20 | centroid shift: 0.288687
Iteration 21 | centroid shift: 0.232586
Iteration 22 | centroid shift: 0.264908
Iteration 23 | centroid shift: 0.274499
Iteration 24 | centroid shift: 0.216477
Iteration 25 | centroid shift: 0.256920
Iteration 26 | centroid shift: 0.268517
Iteration 27 | centroid shift: 0.280223
Iteration 28 | centroid shift: 0.193040
Iteration 29 | centroid shift: 0.060693
Iteration 30 | centroid shift: 0.000000

Converged at iteration 30!
K=6 | WCSS: 169.9309
Initialized 7 centroids from random data points
Centroids shape: (7, 9)
Iteration 1 | centroid shift: 5.790718
Iteration 2 | centroid shift: 1.126239
Iteration 3 | centroid shift: 0.661612
Iteration 4 | centroid shift: 0.617589
Iteration 5 | centroid shift: 0.315154
Iteration 6 | centroid shift: 0.190961
Iteration 7 | centroid shift: 0.357190
Iteration 8 | centroid shift: 0.576489
Iteration 9 | centroid shift: 0.827624
Iteration 10 | centroid shift: 0.577485
Iteration 11 | centroid shift: 0.740011
Iteration 12 | centroid shift: 0.552925
Iteration 13 | centroid shift: 0.702144
Iteration 14 | centroid shift: 0.704168
Iteration 15 | centroid shift: 0.913958
Iteration 16 | centroid shift: 0.827855
Iteration 17 | centroid shift: 0.540032
Iteration 18 | centroid shift: 0.416020
Iteration 19 | centroid shift: 0.289985
Iteration 20 | centroid shift: 0.214084
Iteration 21 | centroid shift: 0.143281
Iteration 22 | centroid shift: 0.117252
Iteration 23 | centroid shift: 0.000000

Converged at iteration 23!
K=7 | WCSS: 160.6675
Initialized 8 centroids from random data points
Centroids shape: (8, 9)
```

```
Iteration 1 | centroid shift: 5.896205
Iteration 2 | centroid shift: 1.369672
Iteration 3 | centroid shift: 0.491559
Iteration 4 | centroid shift: 0.478633
Iteration 5 | centroid shift: 0.527769
Iteration 6 | centroid shift: 0.433060
Iteration 7 | centroid shift: 0.305086
Iteration 8 | centroid shift: 0.343214
Iteration 9 | centroid shift: 0.438289
Iteration 10 | centroid shift: 0.384255
Iteration 11 | centroid shift: 0.418564
Iteration 12 | centroid shift: 0.365610
Iteration 13 | centroid shift: 0.336009
Iteration 14 | centroid shift: 0.524542
Iteration 15 | centroid shift: 0.554315
Iteration 16 | centroid shift: 0.476094
Iteration 17 | centroid shift: 0.311636
Iteration 18 | centroid shift: 0.124542
Iteration 19 | centroid shift: 0.115419
Iteration 20 | centroid shift: 0.162888
Iteration 21 | centroid shift: 0.181345
Iteration 22 | centroid shift: 0.166035
Iteration 23 | centroid shift: 0.206158
Iteration 24 | centroid shift: 0.202236
Iteration 25 | centroid shift: 0.076123
Iteration 26 | centroid shift: 0.000000

Converged at iteration 26!
K=8 | WCSS: 158.4462
Initialized 9 centroids from random data points
Centroids shape: (9, 9)
Iteration 1 | centroid shift: 6.325419
Iteration 2 | centroid shift: 1.569494
Iteration 3 | centroid shift: 0.877290
Iteration 4 | centroid shift: 0.497819
Iteration 5 | centroid shift: 0.375297
Iteration 6 | centroid shift: 0.083582
Iteration 7 | centroid shift: 0.014778
Iteration 8 | centroid shift: 0.000000

Converged at iteration 8!
K=9 | WCSS: 158.1015
Initialized 10 centroids from random data points
Centroids shape: (10, 9)
Iteration 1 | centroid shift: 7.103586
Iteration 2 | centroid shift: 1.759365
Iteration 3 | centroid shift: 0.971347
Iteration 4 | centroid shift: 0.574809
Iteration 5 | centroid shift: 0.263370
Iteration 6 | centroid shift: 0.132026
Iteration 7 | centroid shift: 0.125824
Iteration 8 | centroid shift: 0.062273
Iteration 9 | centroid shift: 0.019486
Iteration 10 | centroid shift: 0.000000
```

```
Converged at iteration 10!
K=10 | WCSS: 154.0722
```



In [12]:
```python
def evaluate_kmeans(labels, y):

    # convert real labels from (2,4) to (0,1)
    # 2 = benign = 0, 4 = malignant = 1
    y_binary = (y == 4).astype(int)

    # try both mappings and pick the better one
    # mapping A: cluster 0 = benign, cluster 1 = malignant
    acc_A = (labels == y_binary).sum() / len(y)

    # mapping B: cluster 0 = malignant, cluster 1 = benign (flipped)
    acc_B = (1 - labels == y_binary).sum() / len(y)

    if acc_A >= acc_B:
        accuracy = acc_A
        final_labels = labels
    else:
        accuracy = acc_B
        final_labels = 1 - labels  # flip the labels

    print(f"K-Means Accuracy: {accuracy * 100:.2f}%")

    # build confusion matrix manually
    TP = ((final_labels == 1) & (y_binary == 1)).sum()  # malignant correctly found
    TN = ((final_labels == 0) & (y_binary == 0)).sum()  # benign correctly found
    FP = ((final_labels == 1) & (y_binary == 0)).sum()  # benign wrongly called mal
    FN = ((final_labels == 0) & (y_binary == 1)).sum()  # malignant wrongly called

    print("\nConfusion Matrix:")
    print(f"                 Predicted Benign    Predicted Malignant")
    print(f"Actual Benign         {TN}                  {FP}")
```

```
    print(f"Actual Malignant      {FN}                  {TP}")

    return accuracy, final_labels


# run it
centroids, labels = kmeans(X, k=2)
accuracy, final_labels = evaluate_kmeans(labels, y)
```

```
Initialized 2 centroids from random data points
Centroids shape: (2, 9)
Iteration 1 | centroid shift: 2.781981
Iteration 2 | centroid shift: 2.840861
Iteration 3 | centroid shift: 0.370358
Iteration 4 | centroid shift: 0.029592
Iteration 5 | centroid shift: 0.014429
Iteration 6 | centroid shift: 0.000000

Converged at iteration 6!
K-Means Accuracy: 95.85%

Confusion Matrix:
                 Predicted Benign    Predicted Malignant
Actual Benign         447                  11
Actual Malignant      18                  223
```

Phase 3 - GMM

In [13]:

```
def initialize_gmm(X, k, kmeans_centroids):

    n_samples, n_features = X.shape

    # means come directly from k-means — this is the key link!
    means = kmeans_centroids.copy()

    # covariances start as identity matrices (one per cluster)
    covariances = []
    for i in range(k):
        cov = np.eye(n_features) * 0.1
        covariances.append(cov)
    covariances = np.array(covariances)

    # weights start equal — each cluster gets same importance
    weights = np.ones(k) / k

    print(f"GMM initialized with K={k}")
    print(f"Means shape      : {means.shape}")
    print(f"Covariances shape: {covariances.shape}")
    print(f"Weights          : {weights}")
    print(f"\nInitial means (from K-Means):")
    for i in range(k):
        print(f"  Cluster {i}: {means[i].round(4)}")

    return means, covariances, weights
```

```
# run it — we pass the centroids we got from kmeans()
means, covariances, weights = initialize_gmm(X, k=2, kmeans_centroids=centroids)
```

```
GMM initialized with K=2
Means shape      : (2, 9)
Covariances shape: (2, 9, 9)
Weights          : [0.5 0.5]

Initial means (from K-Means):
  Cluster 0: [0.227  0.0335 0.0492 0.0375 0.1209 0.033  0.1226 0.028  0.0122]
  Cluster 1: [0.6833 0.642  0.6349 0.5252 0.4953 0.7597 0.5655 0.5641 0.1714]
```

In [14]:

```
def gaussian_pdf(X, mean, cov):

    n_samples, n_features = X.shape

    # add small value to diagonal to prevent singular matrix issues
    cov = cov + np.eye(n_features) * 1e-10

    # compute determinant and inverse of covariance matrix
    det = np.linalg.det(cov)
    inv = np.linalg.inv(cov)

    # normalization constant out front
    norm_const = 1.0 / (((2 * np.pi) ** (n_features / 2)) * (np.sqrt(det)))

    # compute exponent for all points at once
    diff = X - mean  # shape (n_samples, n_features)

    # (x-mu)^T * Sigma^-1 * (x-mu) for each point
    exponent = -0.5 * np.sum(diff @ inv * diff, axis=1)

    probs = norm_const * np.exp(exponent)

    return probs  # shape (n_samples,)


# run it — test on cluster 0
probs = gaussian_pdf(X, means[0], covariances[0])

print(f"Probabilities shape: {probs.shape}")
print(f"First 5 probabilities: {probs[:5]}")
print(f"Min prob: {probs.min():.6f}")
print(f"Max prob: {probs.max():.6f}")
print(f"Any NaN? {np.isnan(probs).any()}")
```

```
Probabilities shape: (699,)
First 5 probabilities: [5.86980791e+00 2.29660871e-03 7.25017746e+00 1.54585219e-03
 5.96643780e+00]
Min prob: 0.000000
Max prob: 7.807619
Any NaN? False
```

In [15]:

```
def e_step(X, means, covariances, weights, k):

    n_samples = X.shape[0]
```

```
    # store probability of each point under each gaussian
    responsibilities = np.zeros((n_samples, k))

    for i in range(k):
        # weight * probability for each cluster
        responsibilities[:, i] = weights[i] * gaussian_pdf(X, means[i], covariances

    # normalize across clusters so each row sums to 1
    row_sums = responsibilities.sum(axis=1, keepdims=True)

    # avoid division by zero
    row_sums = np.where(row_sums == 0, 1e-10, row_sums)

    responsibilities = responsibilities / row_sums

    return responsibilities  # shape (n_samples, k)


# run it
responsibilities = e_step(X, means, covariances, weights, k=2)

print(f"Responsibilities shape: {responsibilities.shape}")
print(f"\nFirst 5 rows (each row should sum to 1):")
print(responsibilities[:5].round(4))
print(f"\nRow sums (should all be 1.0):")
print(responsibilities[:5].sum(axis=1).round(4))
print(f"\nAny NaN? {np.isnan(responsibilities).any()}")
```

```
Responsibilities shape: (699, 2)

First 5 rows (each row should sum to 1):
[[1.     0.    ]
 [0.0088 0.9912]
 [1.     0.    ]
 [0.0079 0.9921]
 [1.     0.    ]]

Row sums (should all be 1.0):
[1. 1. 1. 1. 1.]

Any NaN? False
```

In [16]:

```
def m_step(X, responsibilities, k):

    n_samples, n_features = X.shape

    new_means = []
    new_covariances = []
    new_weights = []

    for i in range(k):

        # responsibilities for cluster i, shape (n_samples,)
        r = responsibilities[:, i]
```

```
        # sum of responsibilities for this cluster
        r_sum = r.sum()

        # update weight
        new_weight = r_sum / n_samples

        # update mean — weighted average of all points
        new_mean = (r[:, np.newaxis] * X).sum(axis=0) / r_sum

        # update covariance — weighted outer product
        diff = X - new_mean
        new_cov = (r[:, np.newaxis] * diff).T @ diff / r_sum

        # add small regularization to keep matrix invertible
        new_cov += np.eye(n_features) * 1e-6

        new_means.append(new_mean)
        new_covariances.append(new_cov)
        new_weights.append(new_weight)

    return np.array(new_means), np.array(new_covariances), np.array(new_weights)


# run it
new_means, new_covariances, new_weights = m_step(X, responsibilities, k=2)

print(f"Updated weights    : {new_weights.round(4)}")
print(f"\nUpdated mean 0: {new_means[0].round(4)}")
print(f"Updated mean 1: {new_means[1].round(4)}")
print(f"\nCovariance 0 shape: {new_covariances[0].shape}")
print(f"Covariance 0 diagonal: {np.diag(new_covariances[0]).round(4)}")
```

```
Updated weights    : [0.668 0.332]

Updated mean 0: [0.2275 0.0344 0.0496 0.038  0.1223 0.038  0.1242 0.0294 0.0114]
Updated mean 1: [0.6861 0.6452 0.639  0.5283 0.4955 0.7556 0.566  0.5655 0.1742]

Covariance 0 shape: (9, 9)
Covariance 0 diagonal: [0.0408 0.0068 0.0107 0.011  0.0092 0.0143 0.0146 0.0106 0.00
61]
```

In [17]:

```
def compute_log_likelihood(X, means, covariances, weights, k):

    n_samples = X.shape[0]
    likelihood = np.zeros(n_samples)

    for i in range(k):
        likelihood += weights[i] * gaussian_pdf(X, means[i], covariances[i])

    # avoid log(0)
    likelihood = np.where(likelihood == 0, 1e-10, likelihood)

    return np.sum(np.log(likelihood))


def gmm(X, k, kmeans_centroids, max_iters=100, tol=1e-4):
```

```
    # step 1 - initialize using k-means results
    means, covariances, weights = initialize_gmm(X, k, kmeans_centroids)

    prev_log_likelihood = None

    for i in range(max_iters):

        # step 2 - e-step
        responsibilities = e_step(X, means, covariances, weights, k)

        # step 3 - m-step
        means, covariances, weights = m_step(X, responsibilities, k)

        # step 4 - check convergence via log-likelihood
        log_likelihood = compute_log_likelihood(X, means, covariances, weights, k)

        print(f"Iteration {i+1} | log-likelihood: {log_likelihood:.4f}")

        if prev_log_likelihood is not None:
            if abs(log_likelihood - prev_log_likelihood) < tol:
                print(f"\nConverged at iteration {i+1}!")
                break

        prev_log_likelihood = log_likelihood

    return means, covariances, weights, responsibilities


# run it
gmm_means, gmm_covs, gmm_weights, gmm_resp = gmm(X, k=2, kmeans_centroids=centroids

print(f"\nFinal weights: {gmm_weights.round(4)}")
print(f"Final mean 0 : {gmm_means[0].round(4)}")
print(f"Final mean 1 : {gmm_means[1].round(4)}")
```

```
GMM initialized with K=2
Means shape      : (2, 9)
Covariances shape: (2, 9, 9)
Weights          : [0.5 0.5]

Initial means (from K-Means):
  Cluster 0: [0.227  0.0335 0.0492 0.0375 0.1209 0.033  0.1226 0.028  0.0122]
  Cluster 1: [0.6833 0.642  0.6349 0.5252 0.4953 0.7597 0.5655 0.5641 0.1714]
Iteration 1 | log-likelihood: 3534.0372
Iteration 2 | log-likelihood: 4377.7732
Iteration 3 | log-likelihood: 4831.2924
Iteration 4 | log-likelihood: 5918.0739
Iteration 5 | log-likelihood: 5977.2466
Iteration 6 | log-likelihood: 6013.8398
Iteration 7 | log-likelihood: 6076.9838
Iteration 8 | log-likelihood: 6103.9694
Iteration 9 | log-likelihood: 6135.3119
Iteration 10 | log-likelihood: 6357.1764
Iteration 11 | log-likelihood: 7183.6968
Iteration 12 | log-likelihood: 7184.8576
Iteration 13 | log-likelihood: 7185.2623
Iteration 14 | log-likelihood: 7185.3392
Iteration 15 | log-likelihood: 7185.3438
Iteration 16 | log-likelihood: 7185.3439
Iteration 17 | log-likelihood: 7185.3439

Converged at iteration 17!

Final weights: [0.5263 0.4737]
Final mean 0 : [0.2037 0.0144 0.0317 0.0217 0.1021 0.     0.1083 0.0057 0.    ]
Final mean 1 : [0.5753 0.4846 0.4826 0.3997 0.4064 0.5832 0.4515 0.4316 0.1383]
```

In [20]:

```python
def evaluate_gmm(responsibilities, y):

    # convert soft responsibilities to hard labels
    labels = np.argmax(responsibilities, axis=1)

    # convert real labels from (2,4) to (0,1)
    y_binary = (y == 4).astype(int)

    # try both mappings
    acc_A = (labels == y_binary).sum() / len(y)
    acc_B = ((1 - labels) == y_binary).sum() / len(y)

    if acc_A >= acc_B:
        accuracy = acc_A
        final_labels = labels
    else:
        accuracy = acc_B
        final_labels = 1 - labels

    print(f"GMM Accuracy: {accuracy * 100:.2f}%")

    # confusion matrix
    TP = ((final_labels == 1) & (y_binary == 1)).sum()
    TN = ((final_labels == 0) & (y_binary == 0)).sum()
```

```python
    FP = ((final_labels == 1) & (y_binary == 0)).sum()
    FN = ((final_labels == 0) & (y_binary == 1)).sum()

    print("\nConfusion Matrix:")
    print(f"                 Predicted Benign    Predicted Malignant")
    print(f"Actual Benign         {TN}                  {FP}")
    print(f"Actual Malignant      {FN}                  {TP}")

    return accuracy, final_labels


# run it
gmm_accuracy, gmm_final_labels = evaluate_gmm(gmm_resp, y)

# compare k-means vs gmm side by side
print("\n****** Final Comparison ******")
print(f"K-Means Accuracy : {accuracy * 100:.2f}%")
print(f"GMM Accuracy     : {gmm_accuracy * 100:.2f}%")

if gmm_accuracy > accuracy:
    print("\nGMM outperformed K-Means! ")
elif gmm_accuracy == accuracy:
    print("\nBoth models performed equally!")
else:
    print("\nK-Means outperformed GMM ")
```

```
GMM Accuracy: 87.12%

Confusion Matrix:
                 Predicted Benign    Predicted Malignant
Actual Benign         368                  90
Actual Malignant      0                   241

****** Final Comparison ******
K-Means Accuracy : 95.85%
GMM Accuracy     : 87.12%

K-Means outperformed GMM
```